

Prime

Description Of Prime

The prime finder is used to identify the prime numbers starting from zero to the specified number of bits given(available spaces) – In this case, we were given a 4-bit input so the available prime numbers are the ones ranging from 0 to 16 {2,3,5,7,11,13}.

Making the circuit

To make the circuit, I first generated the map of possible input and outputs. Image found below. I then generated the k-map to obtain the optimal circuit design (also shown below).

I then found the sum of products and used that for the circuit design.

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	1	1	0
11	1	1	0	1
10	1	0	0	0

$$B'C'D + A'CD + A'B'C'D + AB'D$$

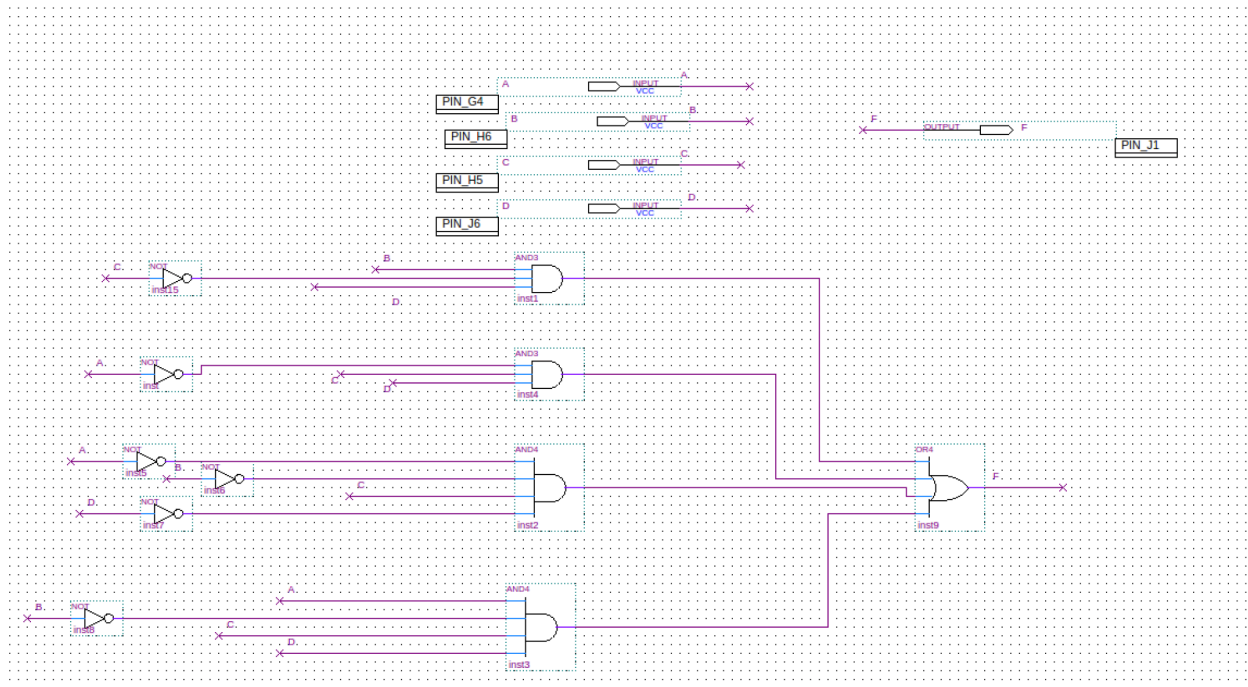


Figure 2.

Description of the results of the testing procedure

The ghdl waveform simulation shows that the circuit works because when you select a prime number (let's say 3 with binary 0011) you can see that the output given by the wave is 1: implying that 3 is indeed a prime number. The same applies to all the prime numbers in this range. Selecting a non-prime yields 0 as expected

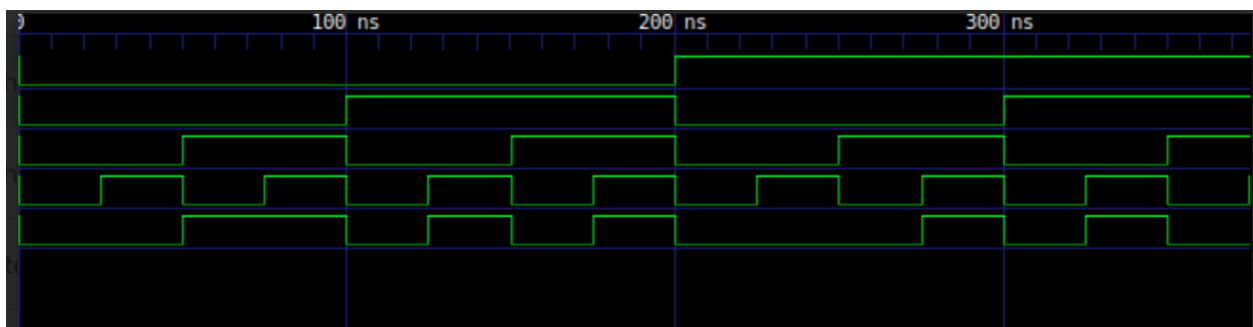


Figure: Picture of ghdl waveform simulation for prime

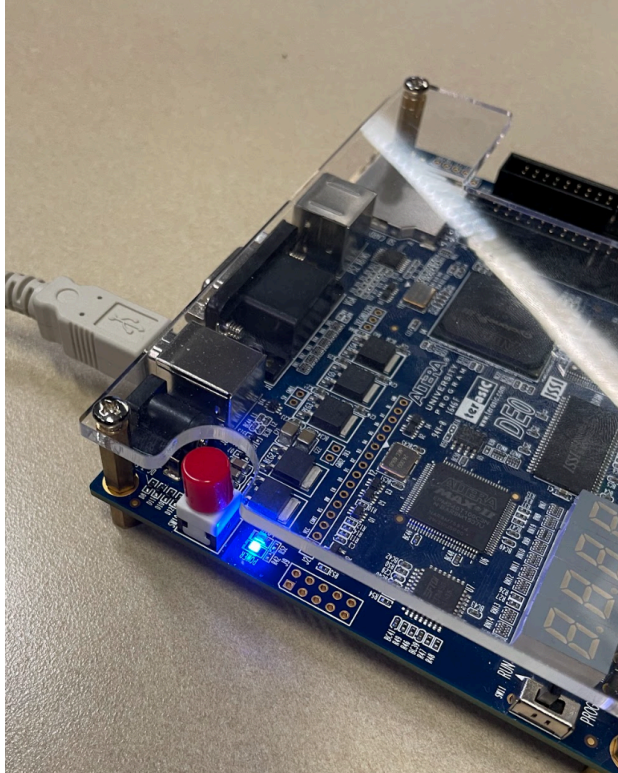


Figure: A video of my prime

Link:

https://drive.google.com/file/d/18vz_rBSCzcKkU9V8HdA0aqP-jDWK52W8/view?usp=sharing

TRAFFIC

Description Of Traffic

This project simulates two stoplights - one controlling north/south traffic and one controlling east/west traffic. The red, green, and yellow lights change as expected when the lights are timed. The simulation has 16 "time steps" (we use a counter to counter from 0 to 15)

Making the circuit

To make the circuit, I generated the respective truth tables for the signals and used those efficient ones obtained after using the k-maps to construct the optimal circuit. The sum of products and some of the k-maps are shown below.

	A	B	C	D	NSR	NSG	NSY	ENR	ENG	ENY
0	0	0	0	0	1	0	0	1	0	0
1	0	0	0	1	0	1	0	1	0	0
2	0	0	1	0	0	1	0	1	0	0
3	0	0	1	1	0	1	0	1	0	0
4	0	1	0	0	0	1	0	1	0	0
5	0	1	0	1	0	1	0	1	0	0
6	0	1	1	0	0	0	1	1	0	0
7	0	1	1	1	0	0	1	1	0	0
8	1	0	0	0	1	0	0	1	0	0
9	1	0	0	1	1	0	0	0	1	0
10	1	0	1	0	1	0	0	0	1	0
11	1	0	1	1	1	0	0	0	1	0
12	1	1	0	0	1	0	0	0	1	0
13	1	1	0	1	1	0	0	0	1	0
14	1	1	1	0	1	0	0	0	0	1
15	1	1	1	1	1	0	0	0	0	1

Figure: Picture showing the truth table for my traffic.

K-MAP FOR THE NORTH/SOUTH

For NSR

AB \ C	00	01	11	10
00	1	0	1	1
01	0	0	1	1
11	0	0	1	1
10	0	0	1	1

For NSG

AB \ C	00	01	11	10
00	0	1	0	0
01	1	1	0	0
11	1	0	0	0
10	1	0	0	0

$$F = A + A'B'C'D'$$

$$F = A'BC' + A'B'C + A'B'C'D$$

For NSY

AB \ C	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	1	0	0
10	0	1	0	0

$$F = A'BC$$

K-MAP FOR THE EAST/WEST

For ENR

AB \ CD	00	01	11	10
00	1	1	0	1
01	1	1	0	0
11	1	1	0	0
10	1	1	0	0

$$F = A' + AB'C'D'$$

For ENG

AB \ CD	00	01	11	10
00	0	0	1	0
01	0	0	1	1
11	0	0	0	1
10	0	0	0	1

$$F = ABC' + AB'C + AB'C'D$$

For ENY

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	1	0
10	0	0	1	0

$$F = ABC$$

Description of the results of the testing procedure

Procedure: After compiling the circuit, I programmed the regulation of the numbers to be controlled by the clock(in this case, a button on the board). This increases the number by 1 whenever it is pressed.

From the attached video, it can be seen that I am regulating the signals with the button. This actually makes sense because when the input is one, we can clearly see that both green lights are on the “RED” (both on the far end). Clicking the button once then changes the one for the N/S(far left in video) to green and leaves the one for the E/W(far right in video) as red.

Successive clicks yield the desired output which concludes that the simulation/circuit works. Clicking the button is a form of increasing the input starting from 1 in intervals of 1.

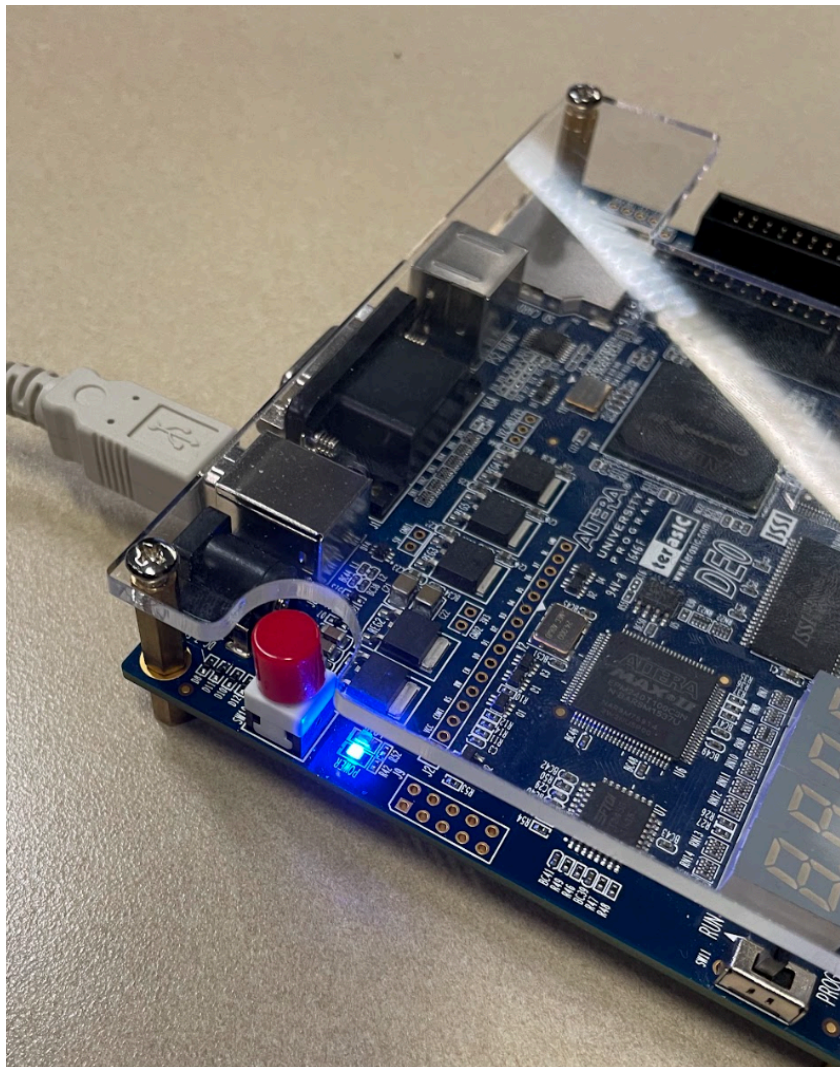


Figure: A video of my traffic

Link:

https://drive.google.com/file/d/1tjEi0BNWP6cytusDTvLJzTbj_dEhzFIb/view?usp=sharing

EXTENSION

Description of the extension

The primeExtension finder is used to identify the prime numbers starting from zero to the specified number of bits given(available spaces). For this extension, I extended the prime number to 6 bits. Therefore, the prime numbers are the ones ranging from 0 to 64 {2,3,5,7,11,13,17,19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61}.

Making the circuit

To make the circuit, I first generated the map of possible input and outputs. A sample of the k-map is provided below, but the most important information(which is the sum of products for This extension file is added below.

Layout

	$\bar{D}\bar{E}\bar{F}$	$\bar{D}E\bar{F}$	$\bar{D}EF$	$D\bar{E}\bar{F}$	$D\bar{E}F$	$DE\bar{F}$	DEF	$D\bar{E}F$
$\bar{A} \cdot \bar{B} \cdot \bar{C}$	0	1	3	2	4	5	7	6
$\bar{A} \cdot \bar{B} \cdot C$	8	9	11	10	12	13	15	14
$\bar{A} \cdot B \cdot \bar{C}$	24	25	27	26	28	29	31	30
$\bar{A} \cdot B \cdot C$	16	17	19	18	20	21	23	22
$A \cdot \bar{B} \cdot \bar{C}$	32	33	35	34	36	37	39	38
$A \cdot \bar{B} \cdot C$	40	41	43	42	44	45	47	46
$A \cdot B \cdot \bar{C}$	56	57	59	58	60	61	63	62
$A \cdot B \cdot C$	48	49	51	50	52	53	55	54

$$F = A'\bar{C}EF + A'B'\bar{C}'DE + A'B'D'EF + A'B'D'E'F + A'B\bar{C}'D'F + A'BDE'F \\ + B\bar{C}DE'F + A\bar{C}'DE'F + AB\bar{C}'D'F + AB'C\bar{E}F + A\bar{C}DE'F$$

Figure: Picture showing the k-map of the primeExtension and the sum of products

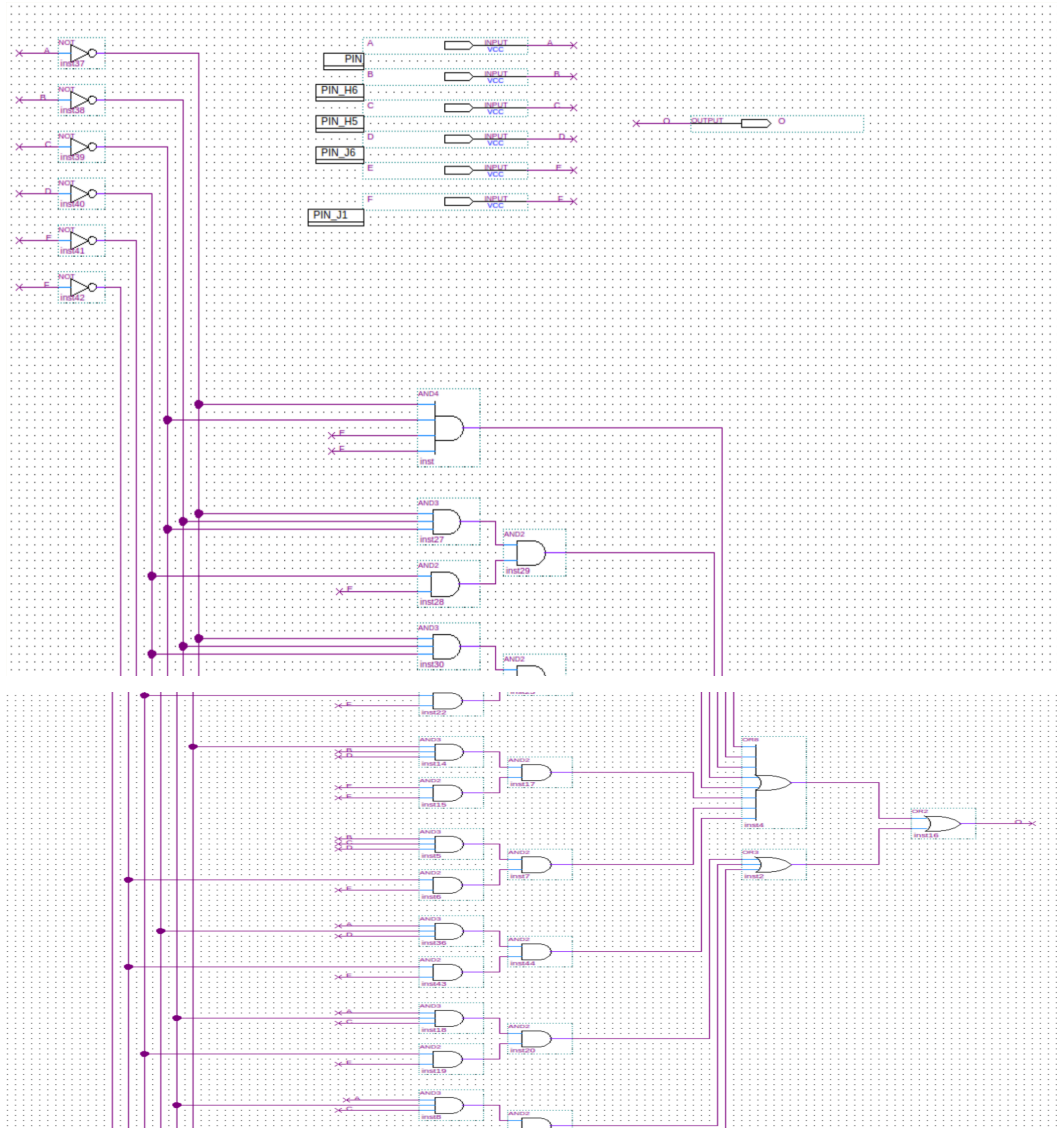


Figure : Picture of the final designed circuit for primeExtension

Description of the results of the testing procedure

I used a similar testing procedure just like the original prime test. The ghdl waveform simulation shows that the circuit works because when you select a prime number (let's say 47 with binary 00101111, you can see that the output given by the wave is 1: implying that 47 is indeed a prime number. The same applies to all the prime numbers in this range. Selecting a non-prime yields 0 as expected.

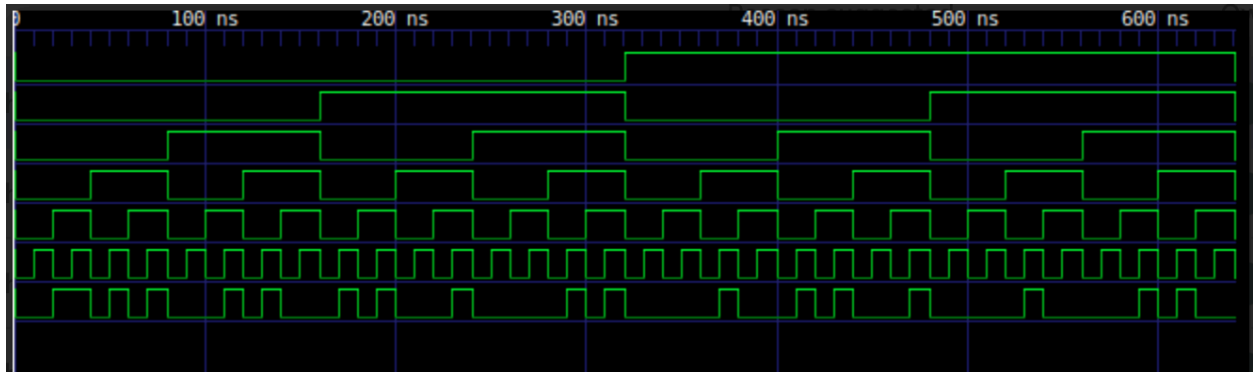


Figure : Picture of ghdl waveform simulation for the primeExtension

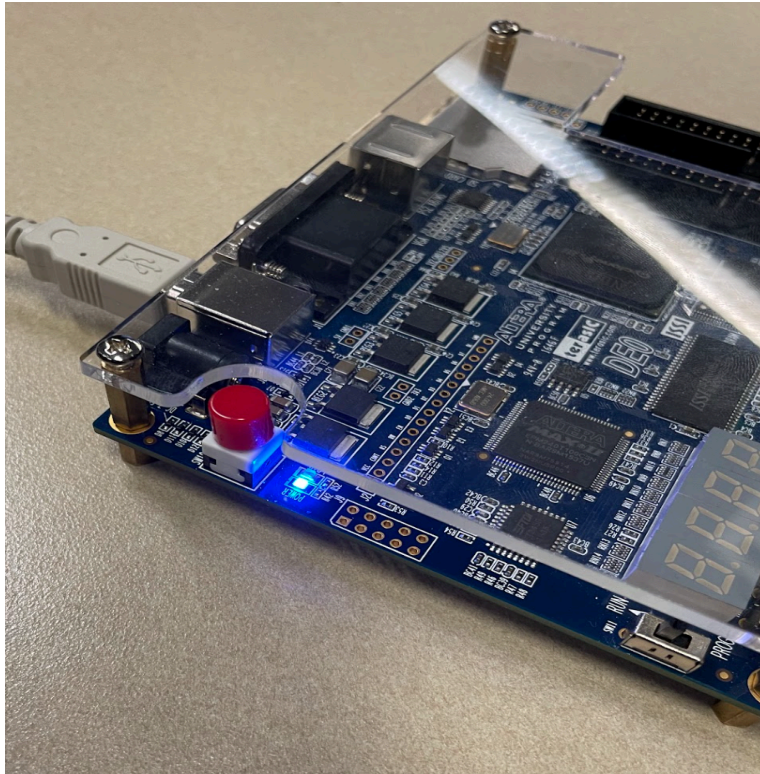


Figure: A video of my extension

Link:

<https://drive.google.com/file/d/1j8KSzFyKceZmGcbfmrCcls-UtkC1At8B/view?usp=sharing>